Step 1: Download NetLogo

Go to:

https://www.netlogo.org/downloads/windows/

Choose:

- **Windows Installer (.exe)** if you use Windows
- **macOS** if you use Mac
- **Linux** version if using Linux

Explore: https://docs.netlogo.org/7.0.4/tutorial1

Step 2: Install

Windows

1. Download the .exe file.
2. Double-click it.
3. Click **Next**.
4. Accept license.
5. Click **Install**.
6. Click **Finish**.

A NetLogo icon will appear on your desktop.

Step 3: Open NetLogo

Double-click **NetLogo**.

You will see:

Interface Tab

Contains:

- Buttons
- Sliders
- Graphics Window
- Monitors

This is where simulations run.

Code Tab

Contains NetLogo code.

Example:

```
to setup
  clear-all
end

to go
end
```

Step 4: Create First Model

Go to **Code** tab.

```
to setup
  clear-all

  create-turtles 10 [
    setxy random-xcor random-ycor
    set color blue
  ]

  reset-ticks
end

to go

  ask turtles [
    rt random 50
    lt random 50
    fd 1
  ]

  tick
end
```

Click **Check**.

If no errors appear, the code is correct.

Step 5: Add Buttons

Go to **Interface** tab.

Add Setup Button

1. Right-click → Add Button
2. Command:

```
setup
```

3. Click OK

Add Go Button

1. Add another Button
2. Command:

```
go
```

3. Tick **Forever**
4. Click OK

Step 6: Run

1. Click **Setup**
2. Click **Go**

You should see turtles moving randomly.

Important Functions for Lab

Function	Purpose
create-turtles	Create agents
ask	Tell agents what to do
fd	Move forward
bk	Move backward

Function	Purpose
rt	Turn right
lt	Turn left
face	Look at target
facexy	Look at coordinates
set color	Change color
die	Remove agent
tick	Advance simulation time
count turtles	Count agents
patch-ahead	Check patch in front
random	Generate random number

Most Common Lab Pattern

```
to setup
  clear-all
  create-turtles 20
  reset-ticks
end

to go
  ask turtles [
    ;; behavior here
  ]
  tick
end
```

- **setup** = create environment and agents
- **go** = simulation loop
- **turtles** = agents
- **patches** = environment
- **tick** = one time step

1. Random movement
2. Move toward target (face)

3. Obstacle avoidance (patch-ahead)
4. Predator-prey (breed)

1. Random Movement

Real-Life Example

A person walking without any destination.

```
ask turtles [
rt random 90
lt random 90
fd 1
]
```

What Happens?

Suppose turtle starts here:

○

After some time:

○ → ↗ → ↓ → ↘ → ↑

It moves randomly in different directions.

Applications

- Animal movement
- Human wandering
- Robot exploration

2. Move Toward Target (face)

Real-Life Example

A food delivery rider moving toward a customer's house.

NetLogo Example

```
ask turtles [
  face patch 10 10
  fd 1
]
```

What Happens?

```
Turtle ---------> Target
```

The turtle always points toward the target and moves closer.

Applications

- GPS navigation
- Delivery robots
- Missiles tracking targets
- Airport passenger movement

3. Obstacle Avoidance (patch-ahead)

Real-Life Example

A robot sees a wall and changes direction.

NetLogo Example

```
ask turtles [

  if [pcolor] of patch-ahead 1 = black [
      rt 90
  ]

  fd 1
]
```

What Happens?

Before:

```
Turtle -----> Wall
```

After:

```
Turtle
   |
   |
   v
```

When it detects a black wall patch ahead, it turns.

Applications

- Self-driving cars
- Cleaning robots
- Drones
- Game AI

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4. Predator–Prey (breed)

Real-Life Example

Wolf chasing sheep.

NetLogo Example

```
breed [wolves wolf]
breed [sheep a-sheep]
```

Create two different agent types.

Sheep Behavior

```
ask sheep [
  rt random 30
  fd 1
]
```

Sheep wander randomly.

Wolf Behavior

```
ask wolves [

  let target one-of sheep

  if target != nobody [
      face target
      fd 1
  ]
]
```

Wolf chases sheep.

Eating Sheep

```
ask wolves [
  ask sheep-here [
      die
  ]
]
```

If wolf reaches sheep:

```
Wolf + Sheep
      ↓
   Sheep dies
```

Applications

- Ecosystem simulation
- Disease spread
- Business competition

- Military simulation

Easy Summary for Viva

Concept	Meaning	Real Example
Random Movement	Agent moves randomly	Person wandering
Move Toward Target	Agent follows a goal	GPS navigation
Obstacle Avoidance	Agent avoids walls	Robot vacuum
Predator-Prey	One agent chases another	Wolf and sheep

Random Movement → Agent moves without a goal.

Move Toward Target → Agent moves toward a specific destination using `face`.

Obstacle Avoidance → Agent checks `patch-ahead` and turns when an obstacle is detected.

Predator-Prey → Different `breed` agents interact, where predators chase and consume prey.

These four examples are the foundation of most Agent-Based Models (ABM) in NetLogo.